

Who to Watch Next: Two-side Interactive Networks for Live Broadcast Recommendation

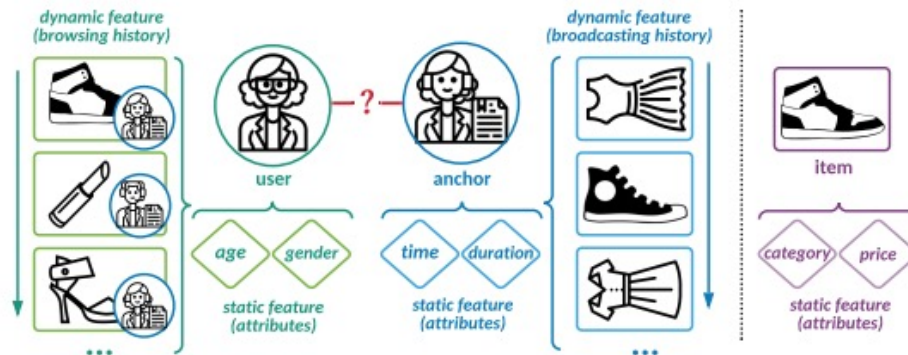
Jiarui Jin*, Xianyu Chen*, Yuanbo Chen*,
Weinan Zhang, Renting Rui, Zaifang Jiang,
Zhewen Su, Yong Yu

Apex Data & Knowledge Management Lab
Shanghai Jiao Tong University



Live Broadcast Recommendation

- Live broadcast recommendation is based on the historical information of both users and anchors, and will automatically generate a list of suitable anchors for users.
- Is the interaction between triple objects (i.e. user, anchor, item), rather than the binary interaction between user and item.



Related Works

- Existing methods in live broadcast recommendation
 - Extracting features manually from the user and anchor side and then feed them into a classifier, which ranges from early matrix decomposition to more recent deep learning.
- Advanced Recommendation Tasks
 - Modifying recommendation tasks: session-based recommendation, intention-based recommendation.
 - Including side information related to users and items in recommendation: studying structural information in user interactions, multimedia-based recommendations.



Related Works

- Collaborative filtering Methods
 - Inner product-based: SVD, SVD++, FM.
 - Deep learning-based: DeepFM, PNN, GraphHINGE
- Two-side sequential networks: exploiting contextual information from both users and items
 - Considering two-side information in an independent fashion
 - Modeling the relationship between all high-order neighbor users and items,



Challenge

- **How to build correlations between users and anchors?**
- How to capture the signal of collaboration between users and anchors?
- How to distinguish key information and filter out the noise?



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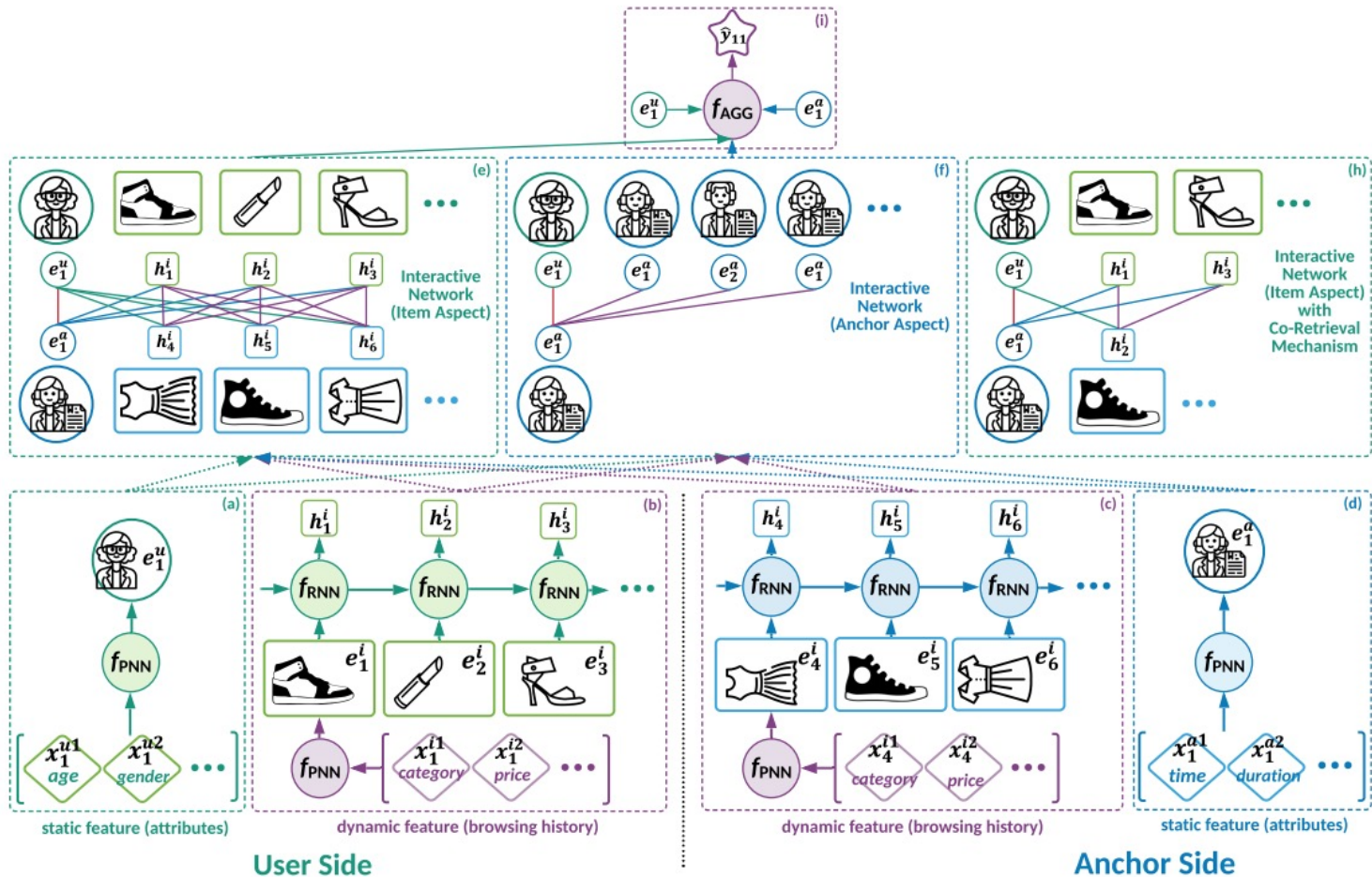


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Twins Framework



Object Modeling

- Introducing PNN to model the static features of each object(i.e., users, anchors, items):

$$e_p^u := f_{\text{PNN}}^u(x_p^u) = v_p^u \odot x_p^u + \sum_{j'=1}^J \sum_{j''=j'+1}^J (v_{j'}^u \odot v_{j''}^u) x_p^{uj'} \cdot x_p^{uj''},$$

- Building sequence models for these dynamic histories of users and anchors, such as RNN models:

$$h_q^i := f_{\text{RNN}}^i(e_q^i | b_{q-1}^i),$$



Interactive Network

- We aim to consider the interaction patterns exploited by the "AND" operation.
- Basic Idea of SVD++:

$$y_{pn}^i = (e_p^u + \sum_{q' \in \mathcal{H}_p^{ui}} \lambda_{pq'} h_{q'}^i)^\top \cdot (e_n^a + \sum_{q'' \in \mathcal{H}_n^{ai}} \beta_{nq''} h_{q''}^i).$$

- An enhanced bi-attention network based on SVD++:

- Item Aspect:

$$\alpha_{pq'nq''}^i = f_{\text{softmax}}^i(\mathbf{w}_{pq'nq''}^{i\top} [e_p^u, h_{q'}^i, e_n^a, h_{q''}^i] + b_{pq'nq''}^i),$$

$$y_{pn}^i = \sum_{q' \in \mathcal{H}_p^{ui}} \sum_{q'' \in \mathcal{H}_n^{ai}} \alpha_{pq'nq''}^i (h_{q''}^i \odot h_{q'}^i),$$

- Anchor Aspect:

$$\alpha_{pn'n}^a = f_{\text{softmax}}^a(\mathbf{w}_{pn'n}^{a\top} [e_p^u, e_{n'}^a, e_n^a] + b_{pn'n}^a),$$

$$y_{pn}^a = \sum_{n' \in \mathcal{H}_p^{ua}} \alpha_{pn'n}^a (e_{n'}^a \odot e_n^a),$$



Co-Retrieval Mechanism

- The time complexity of our interactive network is proportional to the square of the sequence length
- Co-Retrieval Mechanism: find a subset of user's and anchor's related items to feed in the network instead of using the whole data.
- Specific implementation: hard search based on the category of items.

$$\begin{aligned}\widehat{\mathcal{H}}_p^{ui} &= \{i_{q'} | i_{q'} \in \mathcal{H}_p^{ui} \text{ and } x_{q'}^{i1} \in C_{pn}^{ua}\}, \\ \widehat{\mathcal{H}}_n^{ai} &= \{i_{q''} | i_{q''} \in \mathcal{H}_n^{ai} \text{ and } x_{q''}^{i1} \in C_{pn}^{ua}\}.\end{aligned}$$



Optimization Objective

- Direct similarity of users and anchors:

$$y_{pn}^e = e_p^u \odot e_n^a$$

- Aggregate all similarities:

$$\hat{y}_{pn} = \text{sigmoid}(f_{\text{MLP}}([y_{pn}^e, y_{pn}^i, y_{pn}^a])).$$

- Using log loss to optimize parameters:

$$\mathcal{L} = - \sum_{(u_p, a_n) \in \mathcal{D}} (y_{pn} \log \hat{y}_{pn} + (1 - y_{pn}) \log(1 - \hat{y}_{pn})),$$



Experiments

- Datasets
 - Yelp, Trust, Aminer
 - Diantao Live Broadcast Recommendation dataset
- Compared Methods
 - Sequential models: LSTM, NARM, DIN, DIEN, ESMM
 - Interaction models: FM, DeepFM, NeuMF, PNN
 - Sequential model only using the user browsed anchors: X^-
- Our Methods
 - Twins, Twins_i^- , Twins_a^- , Twins_{co}^+
- Evaluation metric: AUC, ACC, LogLoss



Experiments

- Overall Performance

Methods	Yelp Business Dataset			Trust Statement Dataset			Aminer Citation Dataset			Diantao Live Broadcast Dataset		
	LogLoss	ACC	AUC	LogLoss	ACC	AUC	LogLoss	ACC	AUC	LogLoss	ACC	AUC
FM	0.6677	0.5945	0.6233	0.6188	0.7003	0.7574	0.6510	0.6430	0.7071	0.6541	0.6732	0.6896
NeuMF	0.4895	0.7759	0.8424	0.4814	0.7835	0.8440	0.4797	0.7882	0.8542	0.5942	0.7124	0.7229
DeepFM	0.4594	0.7925	0.8658	0.4545	0.8054	0.8756	0.4410	0.8049	0.8826	0.5832	0.7231	0.7345
PNN	0.4581	0.7931	0.8668	0.4452	0.8144	0.8838	0.3932	0.8789	0.9399	0.5432	0.7367	0.7578
DIN ⁻	0.3256	0.8731	0.9420	0.4164	0.8457	0.9154	0.1847	0.9423	0.9799	0.5231	0.7564	0.7790
DIN	0.3156	0.8782	0.9451	0.3771	0.8394	0.9114	0.1212	0.9581	0.9892	0.5100	0.7784	0.7995
LSTM ⁻	0.3236	0.8736	0.9433	0.3931	0.8271	0.9012	0.2218	0.9204	0.9708	0.5334	0.7529	0.7602
LSTM	0.3204	0.8783	0.9445	0.3854	0.8325	0.9051	0.1214	0.9575	0.9891	0.5321	0.7602	0.7789
NARM ⁻	0.3132	0.8811	0.9463	0.3916	0.8306	0.9037	0.2156	0.9216	0.9720	0.5421	0.7432	0.7667
NARM	0.3137	0.8808	0.9463	0.3839	0.8339	0.9060	0.1200	0.9580	0.9893	0.5233	0.7756	0.7953
ESMM ⁻	0.3224	0.8722	0.9414	0.3959	0.8268	0.9008	0.2402	0.9110	0.9655	0.5334	0.7456	0.7698
ESMM	0.3150	0.8776	0.9448	0.4035	0.8262	0.8991	0.1241	0.9564	0.9887	0.5175	0.7753	0.7985
DIEN ⁻	0.3198	0.8723	0.9401	0.4018	0.8388	0.9091	0.2254	0.9183	0.9698	0.5252	0.7657	0.7854
DIEN	0.3291	0.8646	0.9395	0.3911	0.8308	0.9022	0.1242	0.9563	0.9887	0.5145	0.7843	0.8046
TWINS _i ⁻	0.2746	0.8879	0.9538	0.3685	0.8452	0.9174	0.1246	0.9616	0.9893	0.5012	0.7896	0.8010
TWINS _a ⁻	0.2608	0.8948	0.9583	0.3660	0.8446	0.9170	0.1233	0.9622	0.9895	0.4977	0.7920	0.8024
<u>TWINS</u>	0.2603*	0.9120*	0.9659*	0.3528*	0.8501*	0.9235*	0.1081*	0.9631*	0.9913*	0.4855*	0.7934*	0.8187*
TWINS_{co}⁺	0.2596*	0.8962*	0.9593*	0.3579*	0.8458*	0.9194*	0.1170*	0.9612*	0.9903*	0.4731*	0.8045*	0.8205*



Experiments

- Impact of item lists

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- Sequential model V.S. Interaction model

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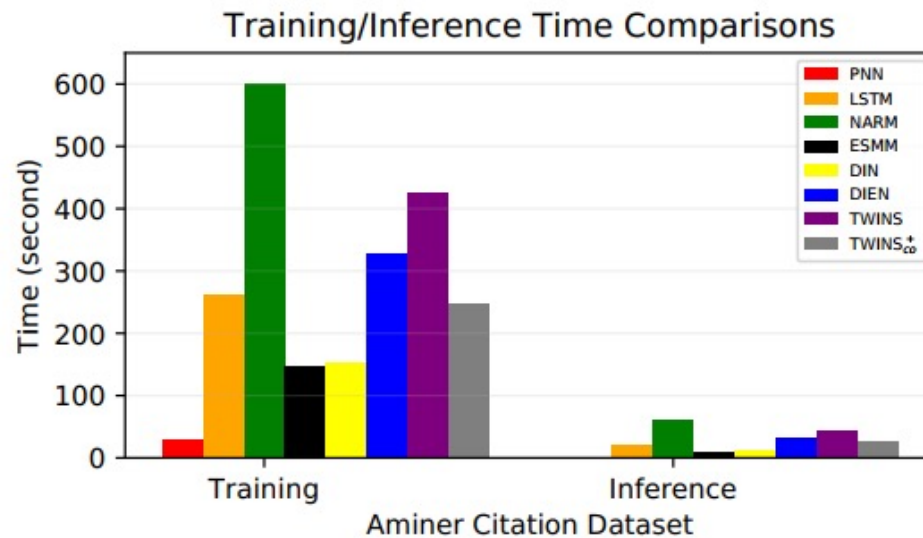
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TWINS _a ⁻	0.2608	0.8948	0.9583	0.3660	0.8446	0.9170	0.1233	0.9622	0.9895	0.4977	0.7920	0.8024
TWINS	0.2603*	0.9120*	0.9659*	0.3528*	0.8501*	0.9235*	0.1081*	0.9631*	0.9913*	0.4855*	0.7934*	0.8187*
TWINS_{co}⁺	0.2596*	0.8962*	0.9593*	0.3579*	0.8458*	0.9194*	0.1170*	0.9612*	0.9903*	0.4731*	0.8045*	0.8205*



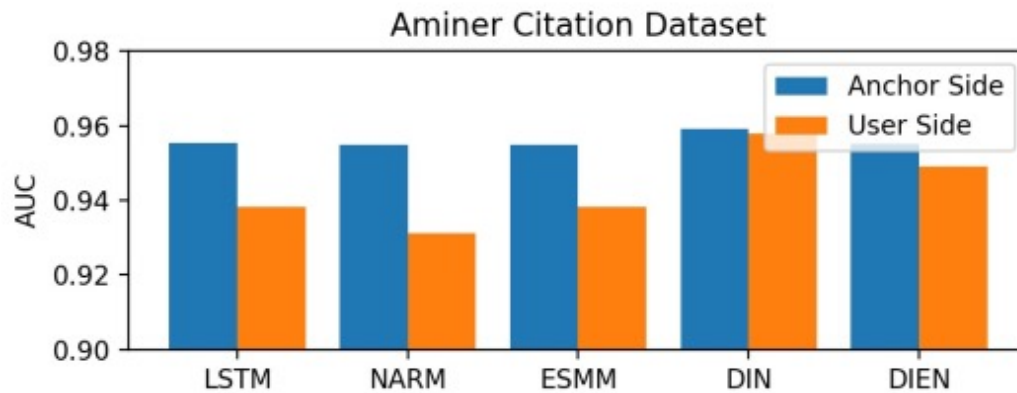
Experiments

- Time Complexity Analysis



Experiments

- Impact of Two-Side Information.



Online Experiments

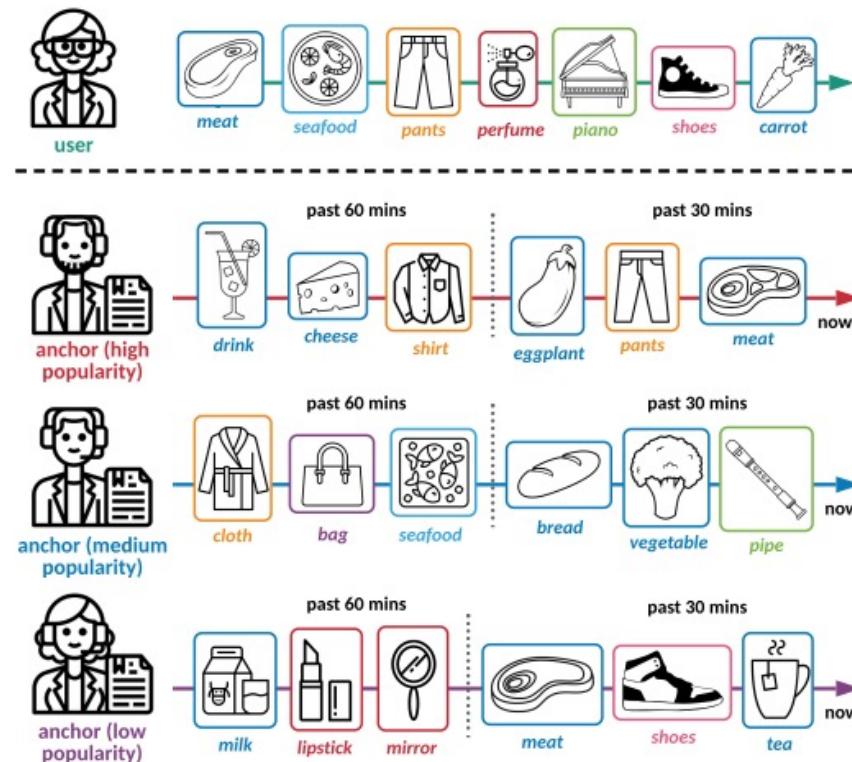
- Improvement of TWINS against current production method on real-world recommendation scenarios.

Recommender	ACTR	UCTR	UCVR
TWINS	8.11%	2.01%	3.52%



Case Study

- A case study to reveal the inner structure of TWINS on Diantao App.



Conclusion

- Investigating a recently emerged live broadcast recommendation and propose a novel two-side framework named TWINS.
- Designing interactive networks from item and anchor aspects to capture the rich interactive patterns in user and anchor sides.
- Developing a co-retrieval mechanism to reduce the high computation costs of the interactive network from the item aspect.



Thanks

Apex Data & Knowledge Management Lab
Shanghai Jiao Tong University

